Journal of
Materials Chemistry C

PAPER

View Article Online
View Journal

Check for updates

Cite this: DOI: 10.1039/c7tc02641b

Spectroscopic signature of moment-dependent electron–phonon coupling in 2*H*-TaS_2

Kapila Wijayaratne,[a] Junjing Zhao,[a] Christos Malliakas,[b] Duck Young Chung,[c] Mercouri G. Kanatzidis[bc] and Utpal Chatterjee *[a]

Received 14th June 2017,
Accepted 6th September 2017

DOI: 10.1039/c7tc02641b

rsc.li/materials-c

Charge density wave (CDW) order is ubiquitous among low-dimensional electronic systems, as realized in layered materials such as various 1*T* and 2*H* polytypes of transition metal dichalcogenides (TMDs). In particular, 2*H*-TaS_2, a prominent member of the vast family of TMDs, hosts a canonical CDW phase transition whose mechanism remains controversial even after decades of research. Using state-of-the-art angle resolved photoemission spectroscopy (ARPES) measurements, we report, for the first time, pronounced many-body renormalizations in 2*H*-TaS_2. These renormalizations are manifested by the presence of multiple slope changes, known as "kinks", in the electronic dispersions. The temperature independence of the kink energies and their close correspondence with the phonon frequencies of the system evidence that these renormalizations are caused by electron–phonon interactions. In addition, the electron–phonon coupling parameter is momentum-dependent, and the CDW vector is not compatible with any of the Fermi surface nesting vectors. Collectively, these observations establish that the origin of the CDW transition in 2*H*-TaS_2 is not the same as that of the Peierls ordering transition, rather this transition is triggered by strong electron–phonon coupling including its momentum anisotropy. The commonality between the ARPES data on 2*H*-TaS_2 and those on 2*H*-$TaSe_2$ and 2*H*-$NbSe_2$ rationalize the above interpretation of the CDW transition being generic to incommensurate CDW orders in 2*H* polytypes of TMDs.

Published on 07 September 2017. Downloaded by Fudan University on 22/09/2017 13:35:38.

1 Introduction

The CDW phase transition of a material spontaneously breaks the symmetry of its nuclear lattice, and a periodic modulation in the electronic charge density occurs at the transition temperature T_{cdw}. The CDW order is believed to be intimately related to some of the most intriguing puzzles in condensed matter physics, such as a pseudogap in cuprate high-temperature superconductors,[1–3] enormous mass renormalization in heavy fermions[4,5] and colossal magnetoresistance in manganites.[6,7] The study of the microscopic details of the CDW transition itself, however, is extremely difficult in the above-mentioned materials, where many other competing interactions are simultaneously present. In contrast, 2*H*-TaS_2, 2*H*-$TaSe_2$ and 2*H*-$NbSe_2$ (hereinafter collectively referred to as 2*H* TMDs) are model systems for such a study—they harbor prototypical CDW orders[8–10] and comprise relatively simple crystal and electronic structures.

Theoretical descriptions of the CDW transition in 2*H* TMDs can be distilled into two broad categories: those mediated by Fermi surface nesting analogous to the Peierls transition in a one-dimensional half-filled system[11] and those mediated by strong electron–phonon coupling.[12–15] If the Peierls picture is applicable, both the real and imaginary parts of the Lindhard susceptibility $\chi_0(\mathbf{q},\bar{\omega})$ should exhibit well-defined peaks[16] at momentum-transfer $\mathbf{q} = \mathbf{q}_{cdw}$ (CDW vector) and energy, referenced to the chemical potential, $\bar{\omega} = 0$. Such peaks are, however, absent in theoretical calculations of $\chi_0(\mathbf{q},\bar{\omega})$.[17] An alternative model relates the CDW orders to the nesting of saddle bands in the vicinity of the chemical potential.[18] But experimentally observed saddle bands do not seem to appear at appropriate energies.[19–25] While these models are in the weak-coupling limit, various experimental findings, for instance the persistence of a non-zero CDW energy gap at temperatures much higher than T_{cdw},[20,21,23] suggest strongly coupled CDWs in 2*H* TMDs. There are indeed a few strong-coupling models,[12–15,26,27] in most of which momentum-dependent electron–phonon coupling is the key to the CDW transition. Several experimental observations are in agreement with these theoretical models.[28–30]

In light of the above discussion, an in-depth study of the electron–phonon coupling in 2*H* TMDs with an emphasis on the momentum ($\mathbf{k}$) anisotropy of the electron–phonon coupling parameter λ is highly desirable for developing a microscopic model of the CDW transition. A straightforward way of

[a] Department of Physics, University of Virginia, Charlottesville, VA 22904, USA. E-mail: uc5j@virginia.edu
[b] Department of Chemistry, Northwestern University, Evanston, IL 60208, USA
[c] Materials Science Division, Argonne National Laboratory, Argonne, IL 60439, USA

Published on 07 September 2017. Downloaded by Fudan University on 22/09/2017 13:35:38.

determining $\lambda(\mathbf{k})$ of a material is to probe the fine structures of its electronic dispersion, obtained from high-resolution ARPES data. Typically, the interaction between electronic excitations and a collective mode such as a phonon or a magnon would modify the bare (*i.e.*, in the absence of interactions) electronic dispersion as well as the electronic lifetime.[31] Defining attributes of the mode are encoded in these modifications, which can be directly inspected by ARPES. Experimental techniques such as tunneling spectroscopy or heat capacity measurements can also investigate the collective modes of a solid. However, these techniques, unlike ARPES, inherently provide Fermi surface averaged information.[32]

Previously, thorough examinations of the impact of collective modes on the electronic excitations *via* ARPES have been conducted on 2*H*-$TaSe_2$ and 2*H*-$NbSe_2$.[33–36] The modes in 2*H*-$NbSe_2$ were inferred to be phonons,[33,36] while the identity of those in 2*H*-$TaSe_2$ could not be resolved.[34,36] Moreover, λ in 2*H*-$NbSe_2$ was found to be momentum-dependent. Surprisingly, such investigations on 2*H*-TaS_2 are absent. In this context, intertwining of superconductivity and CDW at temperatures lower than 0.8 K,[37–39] combined with the influence of relatively strong correlations of Ta d electrons, results in fascinating phase diagrams of 2*H*-TaS_2.[40–49] As in other members of the 2*H* TMDs, the CDW order in 2*H*-TaS_2[50,51] is also incommensurate—the wavelength of the real space charge modulation is not an integer multiple of the lattice constant. T_{cdw} of 2*H*-TaS_2 is $\sim$75 K and $|\mathbf{q}_{cdw}| \sim (2/3 - 0.02)|\overline{\Gamma M}|$.

In contrast to other 2*H* TMDs, ARPES experiments on 2*H*-TaS_2 are limited[19,52,53] and these studies concentrated mostly on the various aspects of the CDW energy gap. The relation between many-body interactions and the CDW order or the connection between $\mathbf{q}_{cdw}$ and the Fermi surface topology remained unexplored. This is precisely the motivation for the current work, in which by using high-resolution ARPES data from 2*H*-TaS_2, we have (i) mapped the minute details of its Fermi surface, and (ii) conducted self-energy analysis inspecting the role of electron–phonon coupling in the CDW transition.

2 Experimental details

We have conducted ARPES measurements using the 21.2 eV helium-I line of a discharge lamp equipped with a Scienta R3000 analyzer at the University of Virginia, as well as 75 eV and 22 eV synchrotron light equipped with a Scienta R4000 analyzer at the Plane Generating Monochromator beamline of the Synchrotron Radiation Center, Wisconsin. The angular resolution is $\sim$0.3 degree, and the energy resolution is $\sim$8–15 meV. For temperature (T)-dependent studies, data were collected in a cyclic way to ensure that there were no aging effects in the spectra. All experiments were performed under ultra-high vacuum (better than 5×10^{-11} Torr, both in the helium lamp system and in the beamline). The single crystal samples were cleaved *in situ* to expose their fresh surfaces for ARPES measurements. The samples were cooled using a closed cycle He refrigerator, and the sample temperatures were monitored using a silicon diode sensor mounted close to the sample holder. During each measurement, the chemical potential was determined by analyzing ARPES data from a polycrystalline gold sample in electrical contact with the sample of interest.[54] Single crystals of 2*H*-TaS_2 were grown using the standard iodine vapor transport method.

3 Results

3.1 Low-energy electronic structures of 2*H*-TaS_2

We start with Fig. 1(a), which depicts the Fermi surface topology of 2*H*-TaS_2 in the normal state. This Fermi surface corresponds to the ARPES data $I(k_x, k_y, \bar{\omega})$ at $\bar{\omega} = 0$ as a function of the in-plane momentum components k_x and k_y. The necessary ARPES data were collected at $T \sim 90$ K $> T_{cdw}$ ($\sim$75 K) with a photon energy of 75 eV. In agreement with first-principles calculations, the Fermi surface of 2*H*-TaS_2 consists of double-walled Fermi surface sheets around the Γ and K points.[55–58] This double-walled nature of the Fermi surface sheets is due to the presence of two formula units per unit cell. In order to construct the Fermi surface intensity map in Fig. 1(a), the pertinent ARPES data have been normalized. The steps in connection with the normalizing procedure are as follows: (i) the background signal, *i.e.*, the constant signal at $\bar{\omega} > 0$, is subtracted from the raw ARPES data; (ii) the area enclosed by the background subtracted

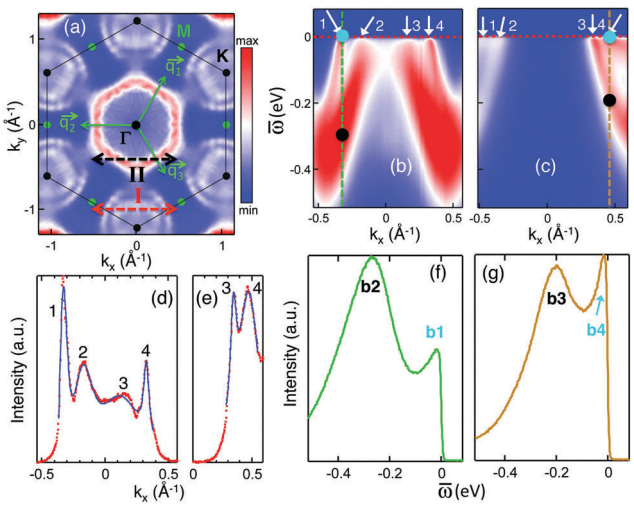

Fig. 1 (a) Fermi surface intensity map of a 2*H*-TaS_2 sample. Green arrows represent the three primary CDW vectors $\mathbf{q}_1$, $\mathbf{q}_2$ and $\mathbf{q}_3$. Energy–momentum intensity map along (b) momentum line I and (c) momentum line II. The momentum lines I and II are marked by the red and black double-sided arrows in (a). Four Fermi crossings in each of (b) and (c) are numbered from 1 to 4 and they are also indicated using white arrows. The red dashed lines in (b) and (c) correspond to the chemical potential. (d) The MDC (shown by red markers) together with the Voigt fit (shown by blue line) at $\bar{\omega} \sim 0$ meV along momentum line I. (e) Same as (d) but along momentum line II. We have focused on the rightmost two peaks (associated with Fermi crossings 3 and 4) in (e) since the intensities of the other two peaks (associated with Fermi crossings 1 and 2) are very weak. (f) The EDC at the momentum location shown by the green dashed line in (b). (g) Same as (f), but at the momentum location shown by the orange dashed line in (c). Double-peak structure of both EDCs signifies bilayer-split bands in (b) and (c).

 This journal is © The Royal Society of Chemistry 2017

Published on 07 September 2017. Downloaded by Fudan University on 22/09/2017 13:35:38.

spectrum and the energy axis between the highest and lowest measured values of $\bar{\omega}$ are calculated; (iii) the intensity of the background subtracted spectrum is divided by the corresponding area (from step (ii)). The raw data covered more than one half of the surface Brillouin zone. For better visualization, the entire Fermi surface intensity map was constructed by reflections, using interpolations to uniform grids. It should be noted that there is no unique normalization procedure for ARPES data as such. The method used here is one among the many such procedures in the literature.[54] We emphasize that the normalization procedure itself has no bearing on our results whatsoever.

The bilayer-split energy bands, which give rise to these double-walled Fermi surface sheets around the Γ and K points, can be better visualized from the ARPES energy–momentum intensity maps in Fig. 1(b) and (c). The intensity maps in Fig. 1(b) and (c) are recorded along momentum lines I and II, respectively. Note that these intensity maps are based on raw ARPES data collected using a photon energy of 22 eV. The Fermi surface and the intensity maps, similar to those seen here, have also been reported for 2*H*-$NbSe_2$ and 2*H*-$TaSe_2$.[19,20,25,33,34]

These bilayer-split bands can also be realized from the one-dimensional representations of ARPES data. For example, ARPES data as a function of either k_x or k_y for a given value of $\bar{\omega}$ is known as the momentum distribution curve (MDC). On the other hand, ARPES data as a function of $\bar{\omega}$ for a specific value of $\mathbf{k}$ is known as an energy distribution curve (EDC). We display a representative MDC in Fig. 1(d) along the momentum line I at $\bar{\omega} \sim 0$ meV. A glance at Fig. 1(b) shows that the momentum line I cuts through four bands at their individual Fermi crossings. Therefore, the MDC in Fig. 1(d) exhibits four peaks. Similar data along the momentum line II have been presented in Fig. 1(e). For visual clarity, we have here focused only on one pair of peaks, since the intensity of the other pair is low. Additionally, two representative EDCs are presented in Fig. 1(f) and (g). The momentum location of the EDC in Fig. 1(f) is indicated by a green dashed line in Fig. 1(b), while that in Fig. 1(g) by an orange dashed line in Fig. 1(b). As expected, each EDC displays a double-peak structure.

Now we turn to the possible interconnection between $\mathbf{q}_{\mathrm{cdw}}$ and Fermi surface nesting vectors. Fig. 1(a) establishes that although the Fermi surface of 2*H*-TaS_2 has a number of nearly parallel regions, their separations do not agree with the magnitude of $\mathbf{q}_{\mathrm{cdw}}$. For instance, both Fermi surface sheets around the Γ point are too large in size to be self-nested by any of the three primary CDW wave vectors $\mathbf{q}_1$, $\mathbf{q}_2$ and $\mathbf{q}_3$ (shown by green arrows in Fig. 1(a)). Therefore, simple Fermi surface nesting is not the driver of the CDW phase transition in 2*H*-TaS_2. A similar conclusion has also been reached in the case of 2*H*-$NbSe_2$.[20,21]

3.2 Multiple kinks in band dispersion

The details of the interaction between a collective mode and the electronic excitations of a system are encapsulated in its single-particle self-energy $\Sigma(\mathbf{k},\bar{\omega})$, a complex-valued function of momentum and energy. The real part $\Sigma'(\mathbf{k},\bar{\omega})$ and the imaginary part $\Sigma''(\mathbf{k},\bar{\omega})$ signify the renormalizations of the bare electronic dispersion and lifetime, respectively.[59]

Both Σ' and Σ'' may, in principle, be evaluated directly from ARPES spectra. This is because the ARPES intensity $I(\mathbf{k},\bar{\omega})$ can approximately be expressed as follows: $I(\mathbf{k},\bar{\omega}) = M(\mathbf{k})A(\mathbf{k},\bar{\omega})f(\bar{\omega})$, where (i) $f(\bar{\omega})$ is the Fermi–Dirac distribution function, (ii) $M(\mathbf{k})$ is the dipole matrix element, (iii) the spectral function $A(\mathbf{k},\bar{\omega}) = \dfrac{\Sigma''(\mathbf{k},\bar{\omega})}{(\bar{\omega} - \varepsilon_{\mathbf{k}} - \Sigma'(\mathbf{k},\bar{\omega}))^2 + \Sigma''(\mathbf{k},\bar{\omega})^2}$ and (iv) $\varepsilon_{\mathbf{k}}$ is the bare electronic dispersion.[54,60–62] For the most generic case, obtaining Σ' and Σ'' from ARPES data is a challenging task since Σ' and Σ'' are the functions of both $\mathbf{k}$ and $\bar{\omega}$. In such a case, one may have to assume some *a priori* representation of the relevant many-body interaction, for making progress in data analysis. However, data analysis gets substantially simplified when Σ' and Σ'' are independent of $\mathbf{k}$ or even have a weak $\mathbf{k}$ dependence.

In the case of $\mathbf{k}$-independent Σ', Σ'', the MDCs at various values of $\bar{\omega}$ take a simple Lorentzian line shape, at least in the vicinity of the Fermi momentum k_{F} such that $\varepsilon_{\mathbf{k}} \sim v_{\mathrm{F}}^0(|k| - k_{\mathrm{F}})$ with v_{F}^0 being the bare Fermi velocity. The Lorentzian line shape of the MDC can be recognized from the above-mentioned form of $A(\mathbf{k},\bar{\omega})$. Consequently, the renormalized dispersion of an energy band can be determined by tracking the fitted peak locations of the corresponding MDCs as a function of $\bar{\omega}$. The deviation of this renormalized dispersion from the bare dispersion provides a measure for $\Sigma'(\bar{\omega})$.[54,60–62] Additionally, $\Sigma''(\bar{\omega})$ can be quantified from the fitted peak widths $W(\bar{\omega})$ of the MDCs. The relation between $\Sigma''(\bar{\omega})$ and $W(\bar{\omega})$ is as follows: $W(\bar{\omega}) = \dfrac{\Sigma''(\bar{\omega})}{v_{\mathrm{F}}^0}$.[54,60–62] One noticeable advantage of this analysis is that no particular model for the many-body interaction is necessary.

The MDCs at $\bar{\omega} \sim 0$ along momentum lines I and II are displayed in Fig. 1(d) and (e) respectively. Notice that the Voigt function instead of the Lorentzian function was used for fitting these MDCs with the aim of approximately taking the momentum resolution into account. We now concentrate on the band dispersion using the MDCs as a function of $\bar{\omega}$. In Fig. 2(a), we show the MDCs together with Voigt fits for a series of $\bar{\omega}$'s along the momentum line I (see Fig. 1(a)). In order to examine how the band dispersion, derived using the MDC analysis, correlates with the corresponding energy–momentum intensity map, we refer to Fig. 2(b). In Fig. 2(b), we superimpose the dispersion curve on the intensity map itself in the vicinity of Fermi crossing 1 (see Fig. 1(b)). Fig. 2(b) zooms into a narrow $\bar{\omega}$–k_x space of the intensity map of Fig. 1(b). This is because the MDC analysis is strictly valid in the vicinity of the Fermi crossing of an energy band. It can be readily observed that the MDC derived band dispersion agrees reasonably well with the intensity profile of the energy–momentum intensity map—at least in an energy window of 100 meV below the chemical potential. A closer look at the dispersion curve in Fig. 2(b) further reveals that the band dispersion consists of multiple changes in slope, commonly referred to as kinks. Typically, the presence of a kink in the electronic dispersion is interpreted as a fingerprint of the scattering of the electrons from some collective mode of the

Published on 07 September 2017. Downloaded by Fudan University on 22/09/2017 13:35:38.

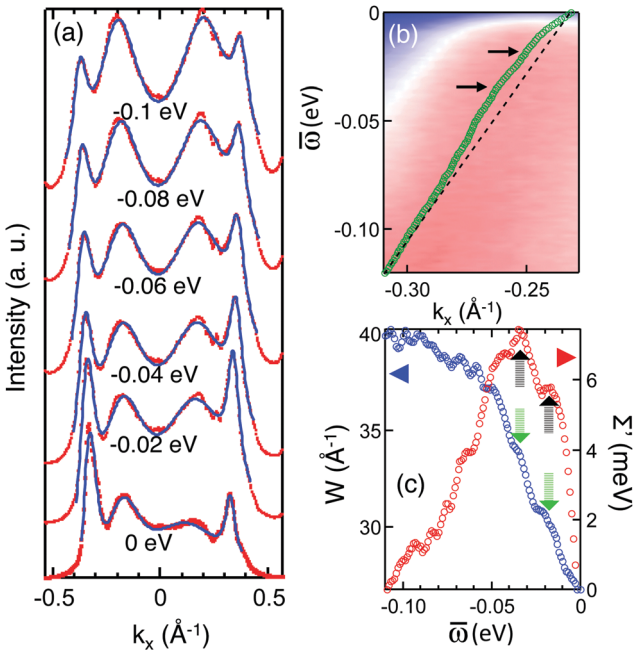

Fig. 2 (a) The MDCs (red markers) together with the fits (blue lines) along the momentum line I (Fig. 1(a)) for a series of $\bar{\omega}$'s. These MDCs are offset for visual clarity. (b) Band dispersion (green circles) from the MDC analysis is superimposed on the energy–momentum intensity map of Fig. 1(b) in the proximity of Fermi crossing 1. An approximated bare band dispersion is shown by a black dashed line, while the kink locations are marked with black arrows in (b). (c) The real part of the self-energy $\Sigma'(\bar{\omega})$ (red circles, right axis) and the width of the MDC peaks $W(\bar{\omega})$ (blue circles, left axis). Note that $\Sigma''(\bar{\omega})$ is directly proportional to $W(\bar{\omega})$. Black dashed arrows indicate the peaks in $\Sigma'(\bar{\omega})$, while the green ones point to the corresponding changes in the slope of $W(\bar{\omega})$. As expected, $\bar{\omega}$ locations of the black and the green dashed arrows approximately coincide.

system. In order to address the identity of this mode, we consulted the published Raman scattering data[63] and theoretical calculations of the phonon spectra[64,65] of 2*H*-TaS_2. The kink energies in the present case are ~18 meV and ~35 meV (shown by black solid arrows in Fig. 2(b)), and they agree well with the phonon frequencies of 2*H*-TaS_2, which are 24.8 meV, 28.2 meV, 34.7 meV and 49.59 meV.[63,64] Therefore, it would be natural to conclude that electron–phonon coupling is responsible for the kinks in the dispersions. It is worth mentioning that similar multiple kinks of phononic origin have been reported in a recent ARPES study of 2*H*-$NbSe_2$.[33] Over the years, kink features in the electronic dispersion, which may or may not be due to phonons, have also been observed in a wide array of solid-state systems other than TMDs, such as metallic systems,[66,67] conventional superconductors,[68] manganites,[69] cuprate high temperature superconductors,[60–62] and pnictide high temperature superconductors.[70]

3.3 $\Sigma'(\omega)$ and $\Sigma''(\omega)$ from MDC analysis

Information on the bare band dispersion is necessary for evaluating $\Sigma'(\bar{\omega})$. We have approximated the bare band dispersion by a straight line, which matches with the high binding energy ($\bar{\omega} \sim -100$ meV) part of the MDC derived dispersion and passes through k_F. A similar approximation for the bare dispersion has been extensively employed in previous ARPES papers.[33,60–62,68–70] We quantify $\Sigma'(\bar{\omega})$ by subtracting the approximated bare band dispersion from the measured one. Additionally, $\Sigma''(\bar{\omega})$ can be obtained from $W(\bar{\omega})$. Both $\Sigma'(\bar{\omega})$ and $W(\bar{\omega})$ are plotted in Fig. 2(c). It is apparent that $\Sigma'(\bar{\omega})$ consists of peaks at $\bar{\omega}$'s close to the kink energies as shown in Fig. 2(b). As expected from the requirements for causality, the peak energies of $\Sigma'(\bar{\omega})$ approximately coincide with the $\bar{\omega}$ locations of the slope changes of $W(\bar{\omega})$.[59] A closer look at Fig. 2(c) reveals that $\Sigma'(\bar{\omega})$ is associated with a number of peaks (energy range: -0.1 eV $< \bar{\omega} < -0.05$ eV) in addition to those we have already taken into account in connection with electron–phonon coupling. The maximum phonon frequency of 2*H*-TaS_2 is 49.59 meV.[63,64] Therefore, any features in $\Sigma'(\bar{\omega})$ beyond this energy are not directly related to electron–phonon coupling. Nevertheless, we speculate the peaks at energies close to -100 meV to be some artifacts of the method for determining $\Sigma'(\bar{\omega})$. However, the peak in the neighborhood of -70 meV can, in principle, correspond to some collective mode. Its robustness and origin will be explored in future work. In passing, we mention that a collective mode of similar energy has been reported for 2*H*-TaS_2, but its origin remains unresolved.[34,36]

3.4 *k*- and *T*-dependence of band renormalization

Now, we illustrate the *k*- and *T*-dependence of the band renormalization. We will first examine the *k*-dependence using Fig. 3(b) and (c). Fig. 3(b) summarizes the band dispersions along the *K*-centered outer Fermi surface sheet, while Fig. 3(c) along the *Γ*-centered inner Fermi surface sheet. As in Fig. 2(b), each dispersion curve in Fig. 3(b) and (c) comprises multiple kink structures. Within experimental error bars, the $\bar{\omega}$ locations of these kinks reside within an energy window of 10–40 meV. This is in accordance with the proposition in Section 3.2—the origin of the kinks in 2*H*-TaS_2 is electron–phonon interactions.

λ can be quantified from the following relation: $\lambda = \frac{v_F^0}{v_F^*} - 1$, where v_F^0 is the bare Fermi velocity, *i.e.*, the slope of the approximated bare dispersion, and v_F^* is the renormalized Fermi velocity, *i.e.*, the slope of the renormalized dispersion at the chemical potential.[60–62] Here the implicit assumption is that the electron–electron interaction does not play a critical role in band renormalization. From Fig. 3(b) and (c), it can be deduced that λ is momentum-dependent along both Fermi surface sheets under current consideration—λ varies from 1.1 to 0.29, which is somewhat weaker in comparison with the momentum variation of λ in 2*H*-$NbSe_2$.[33,35] In future studies, it will be very interesting to explore whether this momentum-anisotropy of λ results in momentum-dependence of the CDW and/or the superconducting energy gap of 2*H*-TaS_2. Furthermore, Fig. 3(d) describes the temperature-dependence of the renormalization effect. It can be seen that the kink structures in the electronic dispersion remains intact even at temperatures higher than T_{cdw} and the kink energies are practically temperature independent. These are consistent with electron–phonon interactions being the origin of the kinks in the electronic dispersions of 2*H*-TaS_2.

 This journal is © The Royal Society of Chemistry 2017

Published on 07 September 2017. Downloaded by Fudan University on 22/09/2017 13:35:38.

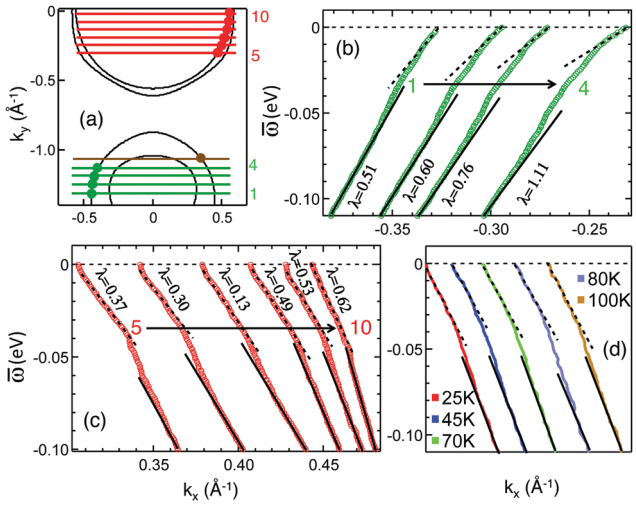

Fig. 3 (a) Schematic plots of Γ- and K-centered Fermi surface sheets. Green and red solid lines show the momentum lines along which the MDCs were constructed for extracting dispersions as shown in (b) and (c). (b) MDC-derived band dispersions close to the Fermi crossings, which are marked by green solid dots and numbered from 1 to 4 in (a). (c) Same as (b) but close to the Fermi crossings, which are marked by red solid dots and numbered from 5 to 10 in (a). The electron–phonon coupling parameter λ at various Fermi crossings are also noted in (b) and (c). (d) MDC-derived band dispersions as a function of T close to the Fermi crossing, which is marked by a brown dot in (a). The brown solid line corresponds to the momentum line along which the pertinent MDCs were constructed. In each dispersion curve, the slope of the black dashed lines passing through the data points at $\bar{\omega}$ = 0 determine v_F*, while the slope of the thick black line measures v_F^0.

4 Conclusions

In summary, our ARPES study of 2H-TaS_2 highlights: (i) the lack of one-to-one correspondence between Fermi surface nesting and CDW vectors, (ii) temperature-independent many-body renormalization mediated by electron–phonon interactions, and (iii) momentum dependence of electron–phonon coupling. From these findings, we conclude that Fermi surface nesting is not relevant to the CDW transition in 2H-TaS_2, rather the origin of the transition is rooted in the momentum-dependent electron–phonon coupling of the system. The similarity between our ARPES data from 2H-TaS_2 and the ones from other 2H TMDs is quite remarkable, which prompts us to conclude that it is the momentum anisotropy of the electron–phonon interaction that determines the characteristics of the incommensurate CDW order in TMDs. Based on the similarity of both the electronic structures and the atomic arrangements of the involved materials, we suggest that this interpretation of the CDW phase transition goes beyond the TMDs, and applies in the same way to other quasi two-dimensional systems, such as high-temperature cuprate superconductors, ruthenates, manganites and cobaltates.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

U. C. acknowledge useful discussions with Patrick Vora and Jasper van Wezel. U. C. acknowledges support from the National Science Foundation under Grant No. DMR-1454304 and from the Jefferson Trust at the University of Virginia. Work at Argonne National Laboratory (C. D. M., D. Y. C., M. G. K.) was supported by the U.S. Department of Energy, Office of Basic Energy Sciences, Division of Materials Science and Engineering.

References

1 J. G. Bednorz and K. A. Müller, *Z. Phys.*, 1986, **64**, 189–193.
2 P. A. Lee, N. Nagaosa and X. G. Wen, *Rev. Mod. Phys.*, 2007, **78**, 17–85.
3 J. Orenstein and A. J. Millis, *Science*, 2000, **288**, 468–474.
4 Theory of Heavy Fermions and Valence Fluctuations, Proceedings of the Eighth Taniguchi Symposium, ed. by Tadao Kasuya and Tetsuro Saso.
5 P. Gegenwart, Q. Si and F. Steglich, *Nat. Phys.*, 2008, **4**, 186.
6 M. B. Salamon and M. Jaime, *Rev. Mod. Phys.*, 2001, **73**, 583–628.
7 E. Dagotto, *New J. Phys.*, 2005, **7**, 67/1–28.
8 J. A. Wilson and A. D. Yoffe, *Adv. Phys.*, 1969, **28**, 193–335.
9 J. A. Wilson, F. J. Di Salvo and S. Mahajan, *Adv. Phys.*, 1975, **24**, 117–201.
10 K. Rossnagel, *J. Phys.: Condens. Matter*, 2011, **23**, 213001.
11 R. Peierls, *More Surprises in Theoretical Physics*, Princeton University Press, Princeton, NJ, 1991.
12 F. Flicker and J. van Wezel, *Nat. Commun.*, 2015, **6**, 7034.
13 F. Flicker and J. van Wezel, *Phys. Rev. B: Condens. Matter Mater. Phys.*, 2016, **94**, 235135.
14 C. M. Varma and A. L. Simons, *Phys. Rev. Lett.*, 1983, **51**, 138–141.
15 L. P. Gor'kov, *Phys. Rev. B: Condens. Matter Mater. Phys.*, 2012, **85**, 165142.
16 G. Gruner, *Density waves in solids*, Perseus Publishing, Cambridge, 1994.
17 M. D. Johannes, I. Mazin and C. A. Howells, *Phys. Rev. B: Condens. Matter Mater. Phys.*, 2006, **73**, 205102.
18 T. M. Rice and G. K. Scott, *Phys. Rev. Lett.*, 1975, **35**, 120–123.
19 W. C. Tonjes, V. A. Greanya, R. Liu, C. G. Olson and P. Molinié, *Phys. Rev. B: Condens. Matter Mater. Phys.*, 2001, **63**, 235101.
20 S. V. Borisenko, A. A. Kordyuk, V. B. Zabolotnyy, D. S. Inosov, D. Evtushinsky, B. B?chner, A. N. Yaresko, A. Varykhalov, R. Follath, W. Eberhardt, L. Patthey and H. Berger, *Phys. Rev. Lett.*, 2009, **102**, 166402.
21 U. Chatterjee, J. Zhao, M. Iavarone, R. Di Capua, J. P. Castellan, G. Karapetrov, C. D. Malliakas, M. G. Kanatzidis, H. Claus, J. P. C. Ruff, F. Weber, J. van Wezel, J. C. Campuzano,

Published on 07 September 2017. Downloaded by Fudan University on 22/09/2017 13:35:38.

R. Osborn, M. Randeria, N. Trivedi, M. R. Norman and S. Rosenkranz, *Nat. Commun.*, 2015, **6**, 6313.
22 K. Rossnagel, O. Seifarth, L. Kipp, M. Skibowski, D. Voß, P. Krüger, A. Mazur and J. Pollmann, *Phys. Rev. B: Condens. Matter Mater. Phys.*, 2001, **64**, 235119.
23 J. Zhao, K. Wijayaratne, A. Butler, J. Yang, C. D. Malliakas, D. Y. Chung, D. Louca, M. G. Kanatzidis, J. van Wezel and U. Chatterjee, *Phys. Rev. B*, 2017, **96**, 125103.
24 R. Liu, W. C. Tonjes, V. A. Greanya, C. G. Olson and R. F. Frindt, *Phys. Rev. B: Condens. Matter Mater. Phys.*, 2000, **61**, 5212.
25 R. Liu, C. G. Olson, W. C. Tonjes and R. F. Frindt, *Phys. Rev. Lett.*, 1998, **80**, 5762.
26 A. Taraphder, S. Koley, N. S. Vidhyadhiraja and M. S. Laad, *Phys. Rev. Lett.*, 2011, **106**, 236405.
27 S. Koley, N. Mohanta and A. Taraphder, *J. Phys.: Condens. Matter*, 2015, **27**, 185601.
28 X. Zhu, Y. Cao, J. Zhang, E. W. Plummer and J. Guo, *Proc. Natl. Acad. Sci. U. S. A.*, 2015, **112**, 2367–2371.
29 F. Weber, S. Rosenkranz, J.-P. Castellan, R. Osborn, R. Hott, R. Heid, K.-P. Bohnen, T. Egami, A. H. Said and D. Reznik, *Phys. Rev. Lett.*, 2011, **107**, 107403.
30 H.-M. Eiter, M. Lavagnini, R. Hackl, E. A. Nowadnick, A. F. Kemper, T. P. Devereaux, J.-H. Chu, J. G. Analytis, I. R. Fisher and L. Degiorgi, *Proc. Natl. Acad. Sci. U. S. A.*, 2012, **110**, 64–69.
31 N. W. Ashcroft and N. D. Mermin, *Solid State Physics*, New York Holt, Rinehart and Winston, 1976.
32 G. Grimvall, *The electron–phonon interaction in metals*, Amsterdam North-Holland Pub. Co., 1981.
33 D. J. Rahn, S. Hellmann, M. Kalläne, C. Sohrt, T. K. Kim, L. Kipp and K. Rossnagel, *Phys. Rev. B: Condens. Matter Mater. Phys.*, 2012, **85**, 224532.
34 K. Rossnagel, E. Rotenberg, H. Koh, N. V. Smith and L. Kipp, *Phys. Rev. B: Condens. Matter Mater. Phys.*, 2005, **72**, 121103(R).
35 T. Valla, A. V. Fedorov, P. D. Johnson, P.-A. Glans, C. McGuinness, K. E. Smith, E. Y. Andrei and H. Berger, *Phys. Rev. Lett.*, 2004, **92**, 086401.
36 T. Valla, A. V. Fedorov, P. D. Johnson, J. Xue, K. E. Smith and F. J. DiSalvo, *Phys. Rev. Lett.*, 2000, **85**, 4759.
37 J. M. E. Harper, T. H. Geballe and F. J. Di Salvo, *Phys. Rev. B: Solid State*, 1977, **15**, 2943.
38 M. H. van Maaren and H. B. Harland, *Phys. Lett.*, 1969, **29A**, 571.
39 S. Nagata, T. Aochi, T. Abe, S. Ebisu, T. Hagino, Y. Seki and K. Tsutsumi, *J. Phys. Chem. Solids*, 1992, **53**, 1259.
40 K. E. Wagner, E. Morosan, Y. S. Hor, J. Tao, Y. Zhu, T. Sanders, T. M. McQueen, H. W. Zandbergen, A. J. Williams, D. V. West and R. J. Cava, *Phys. Rev. B: Condens. Matter Mater. Phys.*, 2008, **78**, 104520.
41 R. M. Fleming and R. V. Coleman, *Phys. Rev. Lett.*, 1975, **34**, 1502.
42 L. J. Li, X. D. Zhu, Y. P. Sun, H. C. Lei, B. S. Wang, S. B. Zhang, X. B. Zhu, Z. R. Yang and W. H. Song, *Phys. C*, 2010, **470**, 313.
43 L. Fang, Y. Wang, P. Y. Zou, L. Tang, Z. Xu, H. Chen, C. Dong, L. Shan and H. H. Wen, *Phys. Rev. B: Condens. Matter Mater. Phys.*, 2005, **72**, 014534.
44 A. Lerf, F. Sernetz, W. Biberacher and R. Schöllhorn, *Mater. Res. Bull.*, 1979, **14**, 797.
45 F. J. Di Salvo, G. W. Hull Jr., L. H. Schwartz, J. M. Voorhoeve and J. V. Waszczak, *J. Chem. Phys.*, 1973, **59**, 1922.
46 F. R. Gamble, J. H. Osiecki and F. J. DiSalvo, *J. Chem. Phys.*, 2003, **55**, 3525.
47 X. Zhu, Y. Sun, S. Zhang, J. Wang, L. Zou, L. E. DeLong, X. Zhu, X. Luo, B. Wang, G. Li, Z. Yang and W. Song, *J. Phys.: Condens. Matter*, 2009, **21**, 145701.
48 D. C. Freitas, P. Rodière, M. R. Osorio, E. Navarro-Moratalla, N. M. Nemes, V. G. Tissen, L. Cario, E. Coronado, M. García-Hernández, S. Vieira, M. Núñez-Regueiro and H. Suderow, *Phys. Rev. B: Condens. Matter Mater. Phys.*, 2016, **93**, 184512.
49 E. Navarro-Moratalla, J. O. Island, S. Mañas-Valero, E. Pinilla-Cienfuegos, A. Castellanos-Gomez, J. Quereda, G. Rubio-Bollinger, L. Chirolli, J. A. Silva-Guillén, N. Agraït, G. A. Steele, F. Guinea, H. S. J. van der Zant and E. Coronado, *Nat. Commun.*, 2016, **7**, 11043.
50 G. A. Scholz, O. Singh, R. F. Frindt and A. E. Curzon, *Solid State Commun.*, 1982, **44**, 1455.
51 J. P. Tidman, O. Singh, A. E. Curzon and R. F. Frindt, *Philos. Mag.*, 1974, **30**, 1191.
52 D. W. Shen, B. P. Xie, J. F. Zhao, L. X. Yang, L. Fang, J. Shi, R. H. He, D. H. Lu, H. H. Wen and D. L. Feng, *Phys. Rev. Lett.*, 2007, **99**, 216404.
53 T. D. Shen, L. Yang, Y. Zhang, J. Shen, L. Fang, J. Yan, D. Ma, H. Wen and D. Feng, *J. Phys. Chem. Solids*, 2008, **69**, 2956.
54 S. Hufner, *Photoelectron Spectroscopy: Principles and Applications*, Springer, USA, 2003.
55 G. Y. Guot and W. Y. Liang, *J. Phys. C: Solid State Phys.*, 1987, **20**, 4315.
56 L. F. Mattheiss, *Phys. Rev. B: Solid State*, 1973, **8**, 3719.
57 N. V. Smith, S. D. Kevan and F. J. DiSalvo, *J. Phys. C: Solid State Phys.*, 1985, **18**, 3175.
58 G. Wexler and A. M. Woolley, *J. Phys. C: Solid State Phys.*, 1976, **9**, 1185.
59 G. D. Mahan, *Many-Particle Physics*, Springer, USA, 2000.
60 J. C. Campuzano, M. R. Norman and M. Randeria, *Physics of Superconductors*, ed. K. H. Bennemann and J. B. Ketterson, Springer, Berlin, 2004, vol. 2, pp. 167–273.
61 A. Damascelli, Z. Hussain and Z. X. Shen, *Rev. Mod. Phys.*, 2003, **75**, 473.
62 P. D. Johnson, A. V. Fedorov and T. Valla, *J. Electron Spectrosc. Relat. Phenom.*, 2001, **117–118**, 153–164.
63 S. Sugai, K. Murase, S. Uchida and S. Tanaka, *Solid State Commun.*, 1981, **40**, 399–401.
64 W. G. Mcmullan and J. C. Irwin, *Can. J. Phys.*, 1984, **62**, 789.
65 P. Lazar, J. Martincova and M. Otyepka, *Phys. Rev. B: Condens. Matter Mater. Phys.*, 2015, **92**, 224104.
66 T. Valla, A. V. Fedorov, P. D. Johnson and S. L. Hulbert, *Phys. Rev. Lett.*, 1999, **83**, 2085.

 This journal is © The Royal Society of Chemistry 2017

67 P. Hofmann, I. Yu Sklyadneva, E. D. L. Rienks and E. V. Chulkov, *New J. Phys.*, 2009, **11**, 125005.
68 D. Mou, R. Jiang, V. Taufour, R. Flint, S. L. Budko, P. C. Canfield, J. S. Wen, Z. J. Xu, G. Gu and A. Kaminski, 2015, arXiv:1503.07069v1.
69 Z. Sun, Y.-D. Chuang, A. V. Fedorov, J. F. Douglas, D. Reznik, F. Weber, N. Aliouane, D. N. Argyriou, H. Zheng, J. F. Mitchell, T. Kimura, Y. Tokura, A. Revcolevschi and D. S. Dessau, *Phys. Rev. Lett.*, 2006, **97**, 056401.
70 L. Wray, D. Qian, D. Hsieh, Y. Xia, L. Li, J. G. Checkelsky, A. Pasupathy, K. K. Gomes, C. V. Parker, A. V. Fedorov, G. F. Chen, J. L. Luo, A. Yazdani, N. P. Ong, N. L. Wang and M. Z. Hasan, *Phys. Rev. B: Condens. Matter Mater. Phys.*, 2008, **78**, 184508.

Published on 07 September 2017. Downloaded by Fudan University on 22/09/2017 13:35:38.